

Housing affordability, Spatial Inequality, and Transport Infrastructure

An empirical analysis of affordability deterioration across London Boroughs(1997-2025)

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Research question: To what extent did transport accessibility and spatial inequality contribute to the outward diffusion of housing affordability of London boroughs between 1997-2005?

1. Introduction:

Housing affordability has increasingly become one of the most significant economic and social challenges affecting London over recent decades. Rising house prices relative to earnings have substantially reduced access to home ownership, particularly amongst younger and lower-income households. Existing research suggests that UK housing affordability has deteriorated considerably since the early 2000s, largely reflecting the interaction between rising property prices, constrained housing supply, and broader structural housing market pressures.

This project investigates changes in housing affordability across London boroughs between 1997 and 2025 using Office for National Statistics (ONS) data. The analysis focuses on borough-level affordability deterioration, spatial inequality, and whether transport-connected boroughs experienced stronger proportional increases in housing unaffordability over time.

The study addresses two principal research questions:

- Did historically lower-income boroughs experience stronger affordability deterioration?
- Did Crossrail-connected boroughs experience significantly greater affordability deterioration relative to non-Crossrail boroughs?

The study addressed two main questions:

1. Did historically lower-income boroughs experience stronger housing affordability deterioration?
2. Did Crossrail-connected boroughs experience greater affordability deterioration than non-Crossrail boroughs?

2. Data and Methodology:

Data

The analysis uses publicly available datasets from the Office of National Statistics (ONS):

- Ratio of House Prices to Earnings by Borough
- Earnings by Workplace, Borough

The affordability is calculated as:

Affordability Ratio = House Price / Annual Earnings

Higher values indicate lower affordability.

The study covers all London boroughs from 1997-2005.

Methodology

The data was cleaned and analysed using R and the tidyverse package program. The percentage change in affordability between 1997-2025 was calculated for each borough using:

Percentage Change = $\frac{\text{Affordability}(2025) - \text{Affordability}(1997)}{\text{Affordability}(1997)} \times 100$

Crossrail-connected boroughs were manually identified and compared against non-Crossrail boroughs using:

- Boxplots
- Descriptive statistics
- Welch two-sample t-tests
- Linear regression models

A regression model was also estimated to examine whether median earnings influenced affordability deterioration.

3. Theoretical Framework and Literature Context:

A central issue within urban economics and public policy debates, lies housing affordability. Contemporary literature suggests that long-run housing affordability pressures are moulded by interactions between housing supply constraints, earnings growth, transport accessibility and spatial-connected dynamics. Widely used, the house price-to-earnings ratio compares property prices relative to local incomes. Recent evidence encapsulates that affordability pressures across the UK remain historically elevated - despite modest improvements, following rising wage growth after 2022.¹

Urban economic literature further suggests that transport accessibility is able to significantly influence housing demand; improving transport infrastructure reduce commuting costs; structurally increasing labour market accessibility; reinforcing an ameliorated transport to generate an "accessibility premium" within connected areas, yet also contributing to rising housing demand and increasing property prices in well-connected locations.

Theoretical notions in relation to bid-rent theory and spatial urban structure further propose that housing demand pressures may diffuse outwards from highly expensive central locations, and towards lower-cost outer areas as households seek relatively affordable alternatives. As a consequence, outer boroughs then may experience stronger proportional affordability deterioration over time due to displacement pressures, gentrification processes and spillover demand effects.²

Previous research examining UK housing cycles also indicates that housing affordability and house prices tend to move in long-term cyclical patterns shaped by broader macroeconomic conditions, credit availability and housing supply constraints.

This project builds on these ideas by assessing whether:

- Transport-connected boroughs experienced stronger affordability deterioration
- Historically lower-income boroughs experienced larger proportional increases in housing unaffordability

¹ Ibid.

² Charalambos Pitros Yusuf Arayici , (2017)," Housing cycles in the UK: a historical and empirical investigation ", Property Management, Vol. 35 Iss 1 pp. 89 - 108

- London's affordability pressures increasingly expanded outwards across the metropolitan area between 1997-2025.

4. Theoretical Expectations:

Contingent with the existing literature prevalent, and spatial housing theory, it was expected that transport-connected boroughs would experience stronger affordability deterioration over time.

Improved transport accessibility can increase the attractiveness of surrounding areas by reducing commuting costs, improving access to employment opportunities.

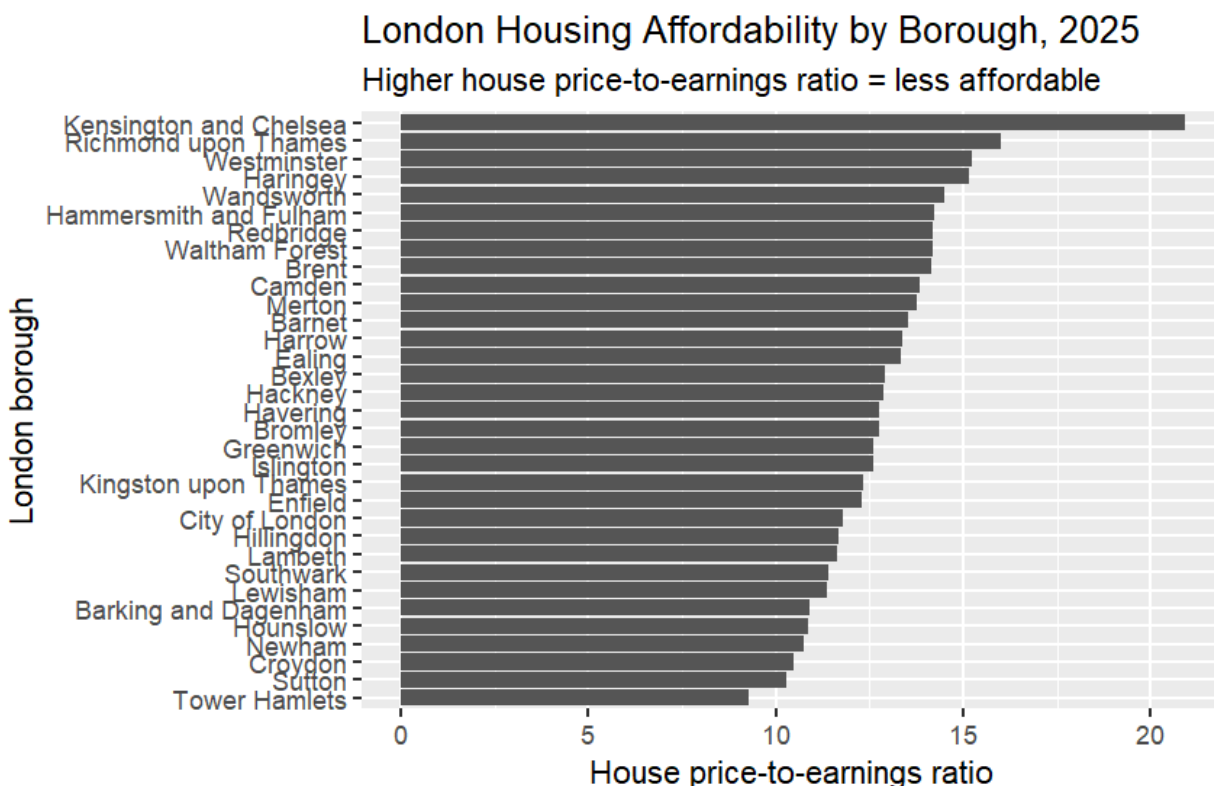
Therefore, it was hypothesised as:

- Cross-rail connected boroughs would then exhibit larger increases in affordability ratios relative to non-connected boroughs
- Lower-income boroughs would experience stronger proportional affordability deterioration over time

The second hypothesis elucidates the possibility that housing demand increasingly diffused outwards from traditionally expensive central boroughs into historically lower-cost outer boroughs. This process may be associated with factors such as:

- Gentrification
- Displacement pressures
- Spatial spillover effects
- Long-run housing supply constraints

The project expected affordability deterioration to increasingly affect outer London rather than remaining concentrated solely within historically expensive London boroughs.



5. Findings & Analysis

Borough-Level Affordability Deterioration

The analysis reveals that housing affordability deteriorated substantially across all London boroughs between 1997 and 2025. The magnitude of deterioration, however, varied considerably across boroughs, suggesting that London's affordability crisis evolved unevenly across the metropolitan area rather than affecting all boroughs uniformly.

Several historically lower-cost outer boroughs experienced some of the largest proportional increases in affordability ratios over the period examined. Boroughs such as Waltham Forest, Newham, Greenwich and Hackney exhibited particularly sharp increases in house price-to-earnings ratios relative to their 1997 baselines. At first glance, this pattern appears broadly consistent with the notion that affordability pressures increasingly diffused outward from traditionally expensive central boroughs into relatively lower-cost outer areas.

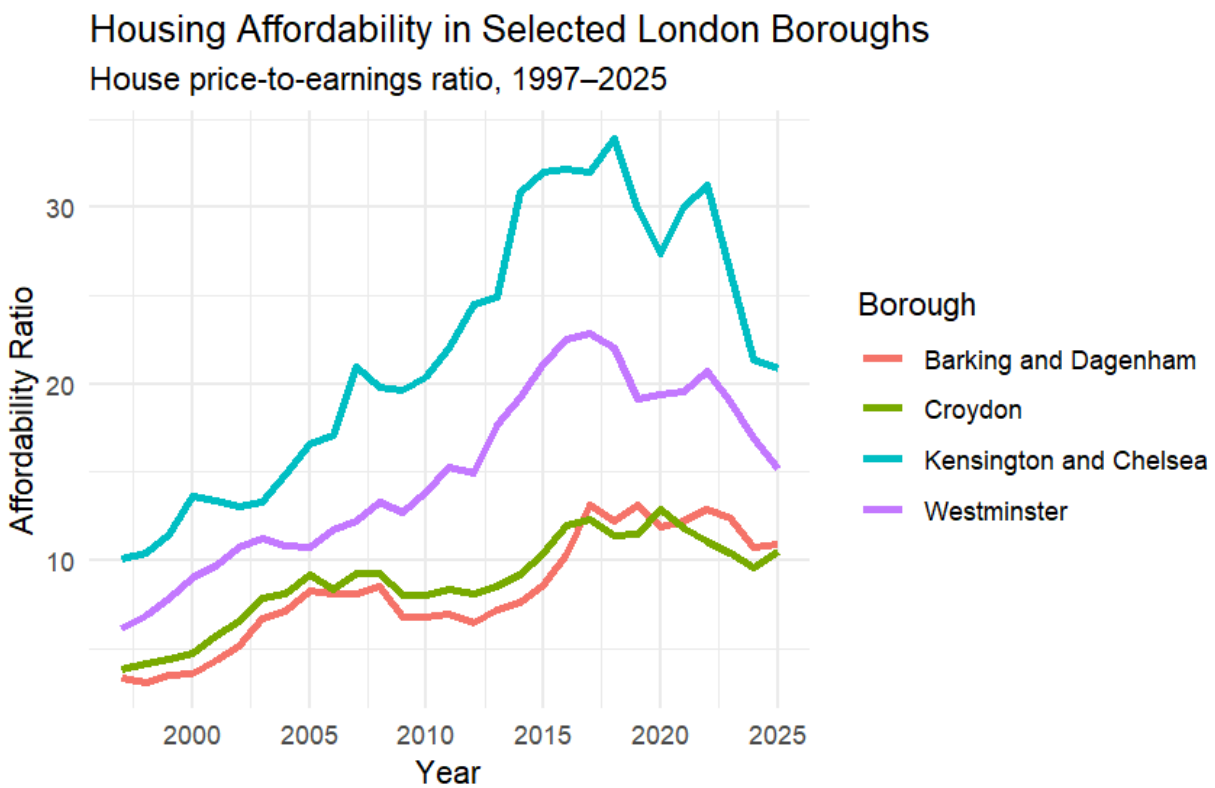
This finding is theoretically significant because it suggests that London's affordability crisis increasingly evolved beyond a geographically concentrated

central London phenomenon. Rather than remaining confined within historically high-value boroughs such as Kensington and Chelsea or Westminster, affordability deterioration increasingly affected outer and historically lower-income boroughs.

One possible interpretation is that escalating central London property values generated outward housing demand spillovers as households sought relatively affordable alternatives beyond traditionally expensive central districts. Consequently, outer boroughs may have experienced disproportionately strong house price growth relative to local earnings, contributing to widening spatial inequality across the metropolitan area.

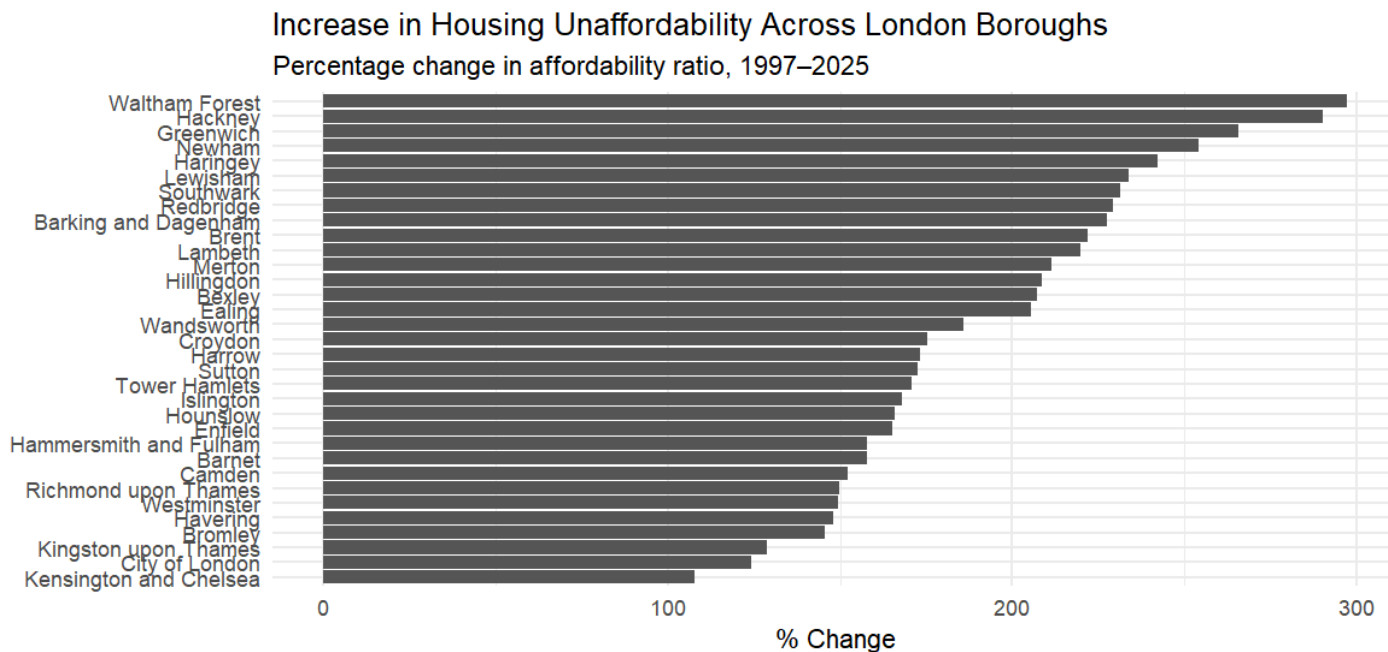
The findings may also reflect broader processes associated with:

- gentrification;
- displacement pressures;
- transport-linked demand spillovers;
- long-run housing supply constraints.



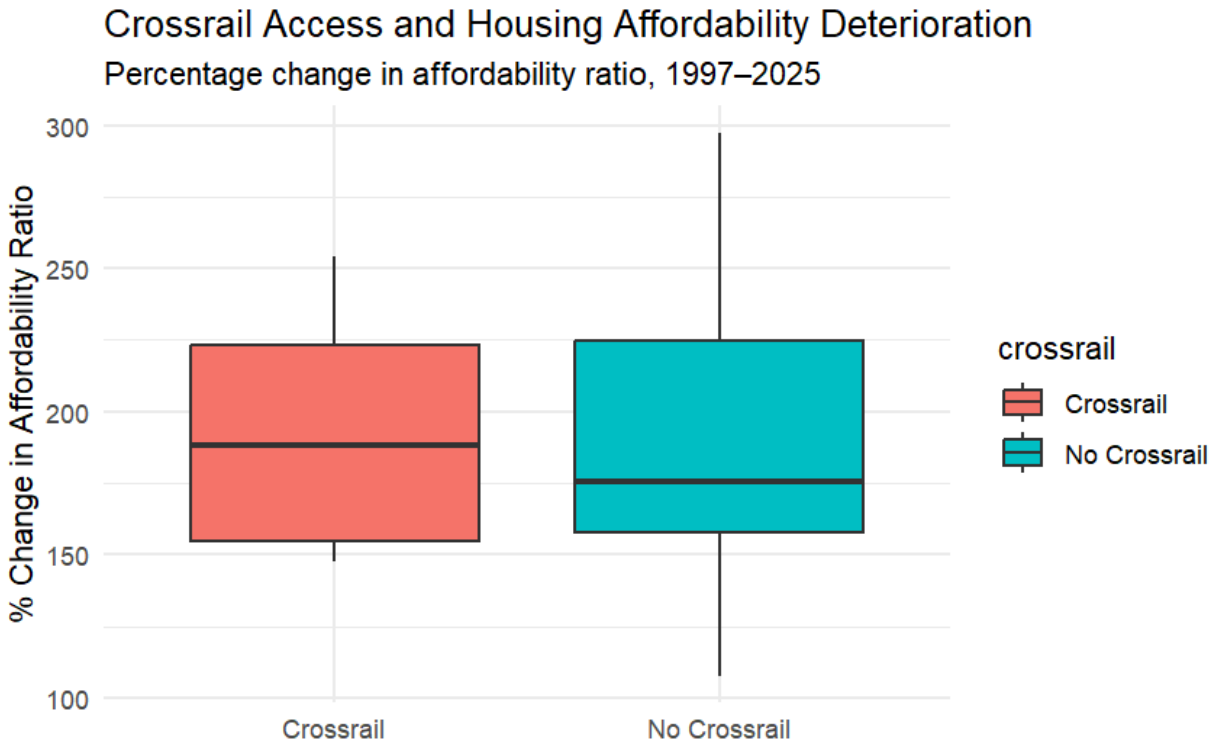
Subsequently, the findings then reinforce the argument that London's housing affordability deterioration has increasingly evolved into a metropolitan-wide phenomenon - rather than just a purely central London issue.

Whilst the descriptive analysis alone cannot establish definitive causal mechanisms, the spatial pattern observed provides preliminary evidence supporting the argument that London's affordability pressures increasingly expanded across the wider urban area between 1997 and 2025.



Crossrail Connectivity and Affordability Deterioration

Drawing upon theories of transport accessibility and urban spatial structure; it was initially hypothesised that Cross-rail connected boroughs would receive the outcome of stronger proportional affordability deterioration, relative to non-connected boroughs. The underlying theoretical explanation denotes urban accessibility models that indicate improvements in transport infrastructure increase local housing demand by reducing commuting costs thereby improving labour market accessibility. In accordance with this, transport investment may also generate localised property price appreciation through an "accessibility premium".



In order to evaluate this relationship empirically, boroughs were categorised into Crossrail-connected and non-Crossrail groups. Comparative analysis was conducted utilising descriptive statistics, boxplot visualisation, and Welch two-sample t-tests.

Conversely, the analysis differentiated from the theoretical explanations made earlier; the study conducted did not identify a statistically significant difference in affordability deterioration between Crossrail and non-Crossrail boroughs.

The Welch two-sample t-test produced:

- P-value = 0.9728

This figure suggests that while transport accessibility may influence localised housing demand and property values, crossrail connectivity independently did not sufficiently explain long-run borough-level affordability deterioration across London during the period examined.

One possible explanation for this is that broader structural housing market dynamics exerted substantially greater influence over affordability outcomes than transport connectivity unaccompanied - this also includes constrained housing supply, population growth and speculative investment.

Moreover, the binary classification of boroughs into Crossrail and non-Crossrail categories may oversimplify the uneven spatial effects of transport infrastructure investment. Particular boroughs may have experienced highly localised price effects concentrated around specific stations rather than borough-wide affordability changes.

Accordingly, the findings presented showcase that transport accessibility should not be interpreted as a singular causal determinant of London's housing affordability crisis.

Regression Analysis: Earnings and Spatial inequality

In further interrogation of the relationship between spatial inequality and affordability deterioration, a multivariable linear regression model was estimated using:

- Crossrail connectivity,
- Median weekly borough earnings

As explanatory variables for percentage affordability deterioration.

The regression analysis identified a statistically significant negative relationship between borough earnings and affordability deterioration. Boroughs with a lower median earnings generally experienced larger proportional increases in affordability ratios over time.

The estimated model was:

$$\overline{\text{Affordability Change}} = 280.89 - 0.112(\text{Median Weekly Earnings})$$

The earnings coefficient was statistically significant at the 5% level:

- $P = 0.0405$

This finding suggests that historically lower-income boroughs experienced stronger relative affordability deterioration between 1997-2025.

A plausible explanation is that escalating housing demand had increasingly diffused outwards from traditionally high-cost central boroughs, and towards historically lower-income outer areas. Thereby, boroughs previously characterised by relative affordability may have experienced disproportionately strong house price growth relative to local changes. The findings are consistent with theories relating to:

- Gentrification;
- Displacement effects;
- Spatial housing demand spillovers;
- Widening metropolitan inequality.

Nevertheless, the regression results must be taken with caution. Although the model identifies statistically significant associations; casual interpretation remains limited due to potential omitted variable bias and broader macroeconomic housing market dynamics not fully captured within the specification.

Factors such as: housing supply elasticity; demographic change; migration patterns; interest rate environments; planning constraints; may also have significantly contributed to affordability outcomes across boroughs during the period examined.

The findings unveil moderate evidence that London's affordability deterioration increasingly affected historically lower-income boroughs - augmenting the argument that housing pressures had expanded spatially across the metropolitan area over time.

6. Discussion and Critical Evaluation

The findings of this project broadly support the notion that London's housing affordability crisis increasingly diffused outward across the metropolitan area between 1997 and 2025, rather than remaining solely concentrated within historically expensive central boroughs. In particular, the statistically significant negative relationship identified between borough earnings and affordability deterioration suggests that historically lower-income boroughs experienced stronger proportional increases in housing unaffordability over time. At first glance, this finding appears broadly consistent with theoretical perspectives relating to bid-rent dynamics and spatial housing demand spillovers, whereby escalating property values within central locations gradually displace demand toward relatively cheaper outer areas.

The interpretation, however, should not be reduced to a singular causal mechanism. Whilst the regression analysis identifies a statistically significant association between lower borough earnings and stronger affordability deterioration, the model itself remains limited in its explanatory scope. Housing affordability outcomes are shaped by a multiplicity of interacting structural pressures, including housing supply constraints, demographic change, migration

flows, speculative investment, planning restrictions, and macroeconomic credit conditions. Consequently, the statistical relationship identified within the model should be interpreted as associative rather than definitively causal.

Moreover, the outward diffusion of affordability pressures may reflect broader processes associated with gentrification and uneven metropolitan development. As affordability deteriorated within traditionally expensive central boroughs, outer boroughs may increasingly have absorbed displaced housing demand from households seeking relatively affordable alternatives. In this respect, affordability deterioration may not simply represent isolated borough-level change, but rather the spatial redistribution of housing pressures across London's wider urban structure.

The absence of a statistically significant Crossrail effect is similarly noteworthy. Urban economics theory often posits that transport accessibility increases local housing demand by reducing commuting costs and improving labour market accessibility, thereby generating an "accessibility premium" within connected areas. On this basis, it was initially hypothesised that Crossrail-connected boroughs would exhibit stronger proportional affordability deterioration relative to non-connected boroughs. The empirical findings, however, did not substantiate this expectation.

Several explanations may account for this discrepancy between theoretical expectation and empirical outcome. Firstly, the effects of transport infrastructure investment may operate at a substantially more localised spatial scale than borough-level aggregation permits. Whilst Crossrail may have increased property values surrounding particular stations and commuter corridors, these effects may not have been sufficiently uniform to materially alter borough-wide affordability trajectories. Consequently, the borough itself may constitute too broad a unit of analysis to fully capture transport-induced housing market effects.

Secondly, broader structural housing market pressures may simply have dominated affordability outcomes across London during the period examined. Long-run housing supply shortages, persistent population growth, speculative housing demand, and broader macroeconomic conditions likely exerted substantially greater influence over affordability trajectories than transport connectivity alone. In this sense, the findings perhaps illustrate the limitations of interpreting transport infrastructure as an independently sufficient explanatory variable within complex metropolitan housing markets.

The methodological limitations of the project should also be acknowledged. Borough-level averages may obscure substantial within-borough inequality, particularly in economically heterogeneous boroughs such as Newham, Westminster, and Tower Hamlets. Additionally, affordability ratios based solely upon house prices relative to earnings do not fully capture broader dimensions of housing pressure, including rental affordability, mortgage financing conditions, household wealth inequality, or interest rate fluctuations.

Nonetheless, despite these limitations, the project provides moderate empirical support for the argument that London's housing affordability crisis increasingly evolved into a broader metropolitan-wide phenomenon disproportionately affecting historically lower-income boroughs. More broadly, the findings demonstrate how quantitative methods and public economic datasets can be utilised to investigate wider questions relating to spatial inequality, urban development, transport accessibility, and the long-run dynamics of metropolitan housing markets.